

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Easy

Question

NASA’s *Cassini* probe has detected an unusual wobble in the rotation of Mimas, Saturn’s smallest moon. Using a computer model to study Mimas’s gravitational interactions with Saturn and tidal forces, geophysicist Alyssa Rhoden and colleagues have proposed that this wobble could be due to a liquid ocean moving beneath the moon’s icy surface. The researchers believe other moons should be examined to see if they too might have oceans hidden beneath their surfaces.

Which choice best states the main idea of the text?

Answer

- A. Rhoden and colleagues were the first to confirm that several of Saturn’s moons contain hidden oceans.
- B. Research has failed to identify signs that there is an ocean hidden beneath the surface of Mimas.
- C. Rhoden and colleagues created a new computer model that identifies moons with hidden oceans without needing to analyze the moons’ rotation.
- D. Research has revealed that an oddity in the rotation of Mimas could be explained by an ocean hidden beneath its surface.

Correct Answer: D

Rationale

Choice D is the best answer. The study isn’t definitive, but it says that Mimas’s wobbly rotation could be explained by the hidden ocean.

Choice A is incorrect. This choice goes too far beyond the information in the text. Rhoden’s team proposed that one moon of Saturn could have a liquid ocean beneath its surface, and that other moons should also be examined, but no one has confirmed anything. Choice B is incorrect. This choice conflicts with the text. Research has identified at least one sign—the unusual wobble in Mimas’s rotation—that might be due to a hidden ocean beneath its surface. Choice C is incorrect. This choice doesn’t reflect the text. The computer model studies “gravitational interactions,” which seem to account for the wobbly rotation of Mimas.